

Exercise 3: Implementing the Builder Pattern

CODE:

```
class Computer {  
    private String cpu;  
    private String ram;  
    private String storage;  
    private String graphicsCard;  
    // Private Constructor  
    private Computer(Builder builder) {  
        this.cpu = builder.cpu;  
        this.ram = builder.ram;  
        this.storage = builder.storage;  
        this.graphicsCard = builder.graphicsCard;  
    }  
    void display() {  
        System.out.println("CPU      : " + cpu);  
        System.out.println("RAM      : " + ram);  
        System.out.println("Storage  : " + storage);  
        System.out.println("Graphics Card : " + graphicsCard);  
        System.out.println();  
    }  
    // Static Nested Builder Class  
    static class Builder {  
        private String cpu;  
        private String ram;  
        private String storage;  
        private String graphicsCard;  
        Builder setCPU(String cpu) {  
            this.cpu = cpu;
```

```

        return this;
    }

    Builder setRAM(String ram) {
        this.ram = ram;
        return this;
    }

    Builder setStorage(String storage) {
        this.storage = storage;
        return this;
    }

    Builder setGraphicsCard(String graphicsCard) {
        this.graphicsCard = graphicsCard;
        return this;
    }

    Computer build() {
        return new Computer(this);
    }
}

public class Main {

    public static void main(String[] args) {

        Computer gamingPC = new Computer.Builder()
            .setCPU("Intel i7")
            .setRAM("16 GB")
            .setStorage("1 TB SSD")
            .setGraphicsCard("NVIDIA RTX 4060")
            .build();

        Computer officePC = new Computer.Builder()
            .setCPU("Intel i5")

```

```
.setRAM("8 GB")  
.setStorage("512 GB SSD")  
.setGraphicsCard("Integrated Graphics")  
.build();  
  
System.out.println("Gaming Computer");  
gamingPC.display();  
  
System.out.println("Office Computer");  
officePC.display();  
  
}  
}
```

OUTPUT:

```
Gaming Computer  
CPU           : Intel i7  
RAM           : 16 GB  
Storage       : 1 TB SSD  
Graphics Card : NVIDIA RTX 4060  
  
Office Computer  
CPU           : Intel i5  
RAM           : 8 GB  
Storage       : 512 GB SSD  
Graphics Card : Integrated Graphics
```